Displaying on-Board Sensor Values on a WebBLE Dashboard

This tutorial shows how to use the web dashboard to read data from the Arduino® Nicla Sense ME sensors.

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Overview

The Arduino® Nicla Sense ME can give you information about the environment such as pressure, temperature and gas readings. Sometimes, you may have to place the sensor in a hard-to-reach area due to certain environmental requirement. Therefore, it will be much convenient and helpful to access the data wirelessly.

Thanks to the ANNA B112 Bluetooth® chip and the libraries developed for the Nicla Sense ME, you can easily stream data over Bluetooth® to a device of your choice. By using WebBLE, no additional software other than a compatible browser (Google Chrome is recommended) is required.

To demonstrate this, we prepared a simple sketch and hosted a dashboard so you can try it yourself

A previous version of this dashboard was developed to be used with the Arduino® Nano 33 BLE. You can see a video that shows how it looks here.

In this tutorial, we will focus on the Arduino® Nicla Sense ME.

Goals

- Upload the sketch to the Arduino® Nicla Sense ME.
- Connect through Bluetooth® Low Energy to our dashboard and read sensor data.

Required Hardware and Software

- Nicla Sense ME
- Micro USB-A cable (USB-A to Micro USB-AB)
- Arduino IDE 1.8.10+, Arduino IDE 2 or Arduino Cloud Editor
- If you choose the Arduino IDE, you will need to install 2 libraries: **Arduino_BHY2** and **ArduinoBLE**

Instructions

Set up the Board

If you use the Cloud Editor to upload the sketch you don't need to install any library. They are all included automatically. If you use the Arduino IDE or the CLI, you need to download the **Arduino_BHY2** and the **ArduinoBLE** libraries.

These libraries can be found within

IDE, or it can be downloaded separately following the links attached within required hardware and software section.

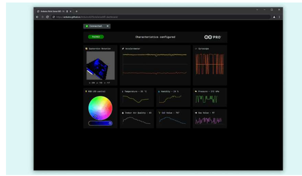
If you use a local IDE, you can copy & paste the following sketch:

```
/*
Arduino Nicla Sense ME
Hardware required: htt
1) Upload this sketch
2) Open the following
https://arduino.githut
3) Click on the green
Web dashboard by D. Pa
Device sketch based or
Sketch and web dashboa
*/

#include "Nicla_Systen
#include "Arduino_BHY2
#include <ArduinoBLE.I

#define BLE_SENSE_UUII

const int VERSION = 0x

BLEService service(BLE

BLEUnsignedIntCharacte
BLEFloatCharacteristic
BLEUnsignedIntCharacte
BLEFloatCharacteristic

BLECharacteristic acce
```

Once you have these tools, you can select the Nicla Sense ME as target board and its corresponding port. Now you are ready to upload the sketch.

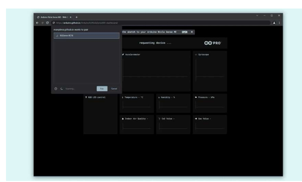
Connect to the Dashboard

Once you uploaded the sketch to your board you can open the Nicla Sense ME Dashboard. If you're interested in the source code, you can have a look at the repository.



Preview of the dashboard

To connect your board to the dashboard, you will need to click on the top left button which says "Connect Board". A pop up will be displayed in your browser and it starts searching for Bluetooth® devices. This application leverages the WebBLE functionality of your browser.



Popup message to connect the device to the browser

> For this feature to work, make sure that WebBLE is both supported and enabled! In Google Chrome go to chrome://flags and enable "Experimental Web Platform features". Check the website compatibility list to confirm that your browser supports this feature

Once it is connected, the button will change its color to green, and the graphs will start to show data in real time. You will be able to verify its operation by trying out the following actions:

Try to rotate the board and see the 3D model of the board spin.

You can also select a different LED color from the bottom left widget.

Breathe onto the board and see the humidity and temperature values changing.

Conclusion

The Nicla Sense ME supports a lot of use cases through its on-board sensors and the Bluetooth® Low Energy functionality. By leveraging the WebBLE API, you do not need to install or run any application from your computer as shown in this tutorial. You can read more about WebBLE technology **here**.